Mathematical Statistics (I)
Midterm Exam.

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1.(40) Let X be a r.v. with mgf $M(t) = (1-t)^{-3}, t < 1$. Find $E(X^k)$ for some positive integer k.

2.(60) Let X be a r.v. with mgf $M(t), t > 0$. Show that $P(X \geq a) \leq e^{-at}M(t)$.

3.(180) Let (X, Y) have the joint pdf $f(x, y) = 8xyI(0 < x < y < 1)$, and let $Z = X/Y$ and $W = Y$.
(1)(60) Find the cdf of Z.
(2)(40) Using the transformation technique, find the jpdf of (Z, W).
(3)(40) Are X and Y independent? Why or why not?
(4)(40) Are Z and W independent? Why or why not?

4.(60) Let $f(x, y) = (x + y)I(0 < x < 1)I(0 < y < 1)$ be the jpdf of (X, Y). Compute the correlation coefficient between X and Y.

5.(160) Let (X, Y) have the joint pdf $f(x, y) = 2I(0 < x < y < 1)$.
(1)(40) Compute $E(X|Y)$.
(2)(40) Compute $Var(E(X|Y))$.
(3)(40) Compute $Var(X|Y)$.
(4)(40) Compute $E(Var(X|Y))$.